

Kazi Hafiz Md. Asad

Efficient NLP · LLM Compression · Higher-Order Machine Learning

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Portfolio | Google Scholar | GitHub | LinkedIn | LeetCode | Deep-ML

3 publications (1 Q1 journal, 1 IEEE conference, 1 preprint under review) | **29M-parameter** distilled BERT student at **3.8×** compression | **7B-parameter** LLM fine-tuned under constrained VRAM across **27** controlled runs | Leakage-controlled benchmark over **9** interaction datasets

Research Profile

M.S. researcher in efficient and trustworthy machine learning, working where model compression meets rigorous empirical evaluation. My work asks a single recurring question: *how much capacity does a model actually need, and how do we prove it?* I have answered it in three settings, knowledge distillation of BERT into a 29M-parameter student that retains task performance at a fraction of the cost (*PLOS ONE*, 2025), resource-matched comparison of classical and transformer text classifiers (*IEEE QPAIN*, 2025), and constrained-VRAM fine-tuning of a 7B-parameter LLM through dynamic progressive compression (M.S. thesis, in progress). A parallel line of work extends the same evidentiary standard to structure learning, testing when hypergraph representations genuinely beat pairwise graphs for protein-complex recovery under strict leakage control. Across every project I build the baseline before the method, run the ablation before the claim, and report the efficiency-quality trade-off in full.

Research interests: Efficient and trustworthy NLP; parameter-efficient LLM adaptation; quantisation, pruning, and distillation; higher-order and graph representation learning; computational biology; sustainable AI; multimodal and longitudinal modelling.

What I bring to a lab: Independent experimental design and reproducible engineering; first-author manuscript writing; comfort with GPU-constrained research; a record of collaborating across NLP, network science, and computational biology.

Education

M.S. in Computer Science and Engineering

North South University

Jul 2024 – Present

Dhaka, Bangladesh

GPA 3.55/4.00; 24 of 30 credits completed. Supervisor: Dr. Mohammad Ashrafuzzaman Khan.

Thesis: *Ultra-Efficient Fine-Tuning of 7B-Parameter Large Language Models under Constrained VRAM using Dynamic Progressive Compression and Adaptive Stride Training.*

B.S. in Computer Science and Engineering

North South University

May 2019 – Jun 2023

Dhaka, Bangladesh

GPA 3.53/4.00; graduated **Cum Laude**.

Thesis: *BERT Knowledge Distillation for NLP Tasks* - extended into the 2025 *PLOS ONE* article below.

Graduate coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Advanced Algorithms, Graph Theory, Digital Image Processing, Computer Security, Probability and Statistics, Databases, Software Engineering, Advanced VLSI Design.

Publications

2025. Ahmed Akib Jawad Karim, **Kazi Hafiz Md. Asad**, and Md. Golam Rabiul Alam. “**Larger models yield better results? Streamlined severity classification of ADHD-related concerns using BERT-based knowledge distillation.**” *PLOS ONE*, 20(2), e0315829. [Q1 journal] DOI | arXiv

- ▶ Introduced LASTBERT, a 29M-parameter student distilled from BERT-base (110M) - a **3.8×** parameter reduction - and showed it holds competitive accuracy across GLUE tasks and on ADHD concern-severity classification from social-media text.
- ▶ Led formal analysis, validation, and visualisation; designed the comparative evaluation that grounds the paper’s central claim that scale is not a prerequisite for task competence.

2025. Ahmed Akib Jawad Karim, **Kazi Hafiz Md. Asad**, and Aznur Azam. “**Strengthening False Information Propagation Detection: Leveraging SVM and Sophisticated Text Vectorization Techniques in Comparison to BERT.**” *2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pp. 1–6. [IEEE] DOI | arXiv

- ▶ Benchmarked BERT against SVM classifiers over TF-IDF, Word2Vec, and bag-of-words representations for fake-news detection under matched resource budgets.
- ▶ Established that a carefully specified classical pipeline approaches transformer-level performance at a small fraction of the compute a practical result for low-resource deployment.

2026. Kazi Hafiz Md. Asad, Rafi Majid, Md Tanjeelur Rahman Labib, and Ahsanur Rahman. “**When Does Higher-Order Representation Improve Protein-Complex Recovery? A Leakage-Controlled, Matched- k Comparison of Graph and Penalized Hypergraph Spectral Clustering.**” *Preprint, under review.* [first author] [Manuscript](#)

- ▶ Designed and built a leakage-free benchmark across **9** IntAct MITAB datasets with held-out evaluation, matched- k comparison, significance testing, null models, and Gene Ontology enrichment.
- ▶ Characterised the empirical regime in which higher-order incidence structure outperforms pairwise graph projection - converting a widely assumed advantage into a conditional, measured one.

Research Experience

M.S. Thesis Researcher

North South University : Dept. of Electrical and Computer Engineering

2024 – Present

Dhaka, Bangladesh

Ultra-Efficient Fine-Tuning of 7B-Parameter LLMs under Constrained VRAM

- ▶ Built a memory-efficient framework that fine-tunes Qwen2.5-7B-Instruct on a single consumer-class GPU, combining 4-bit quantised loading, QLoRA, gradient checkpointing, adaptive rank scheduling with SVD warm-start, gradient-noise-gated control, and GPU isolation.
- ▶ Ran **27** controlled experiments measuring model quality, peak GPU memory, wall-clock runtime, convergence stability, and sensitivity to the compression schedule; isolated which components carry the gains and which are redundant.
- ▶ Engineered the codebase for reproducibility end to end: pinned environments, controlled seeds, full configuration logging, and per-experiment result tracking.

Research Collaborator : Efficient NLP Group

North South University

2023 – Present

Dhaka, Bangladesh

- ▶ Research with [Dr. Mohammad Ashraful Khan](#) on knowledge distillation, efficient NLP, experimental validation, and scientific writing.
- ▶ Contributed to two peer-reviewed papers spanning transformer compression and computationally efficient misinformation detection, from experiment design through camera-ready submission.

Manuscripts and Reviews in Preparation

Graph vs. Hypergraph Protein-Complex Recovery – *network science and computational biology*

[Code](#)

- ▶ Modelled multi-protein experimental interactions as both co-complex graphs and incidence-based hypergraphs; proposed a tunable penalised hypergraph Laplacian and compared spectral clustering over CORUM and 9 MITAB datasets under strict leakage control.

Sustainable Intelligence – *environmental trade-offs of cloud-based machine learning*

- ▶ PRISMA-guided critical review of energy, carbon, water, and hardware-lifecycle impacts across training, inference, and data-centre operation; assessed measurement inconsistency and mitigation via carbon-aware scheduling, algorithmic efficiency, and low-carbon deployment.

Naturalistic Behavioural Prognosis in Autistic Children – *digital phenotyping and developmental forecasting*

- ▶ Synthesised longitudinal evidence from naturalistic audio, video, wearable, and behavioural data for forecasting language and adaptive-functioning outcomes; identified gaps in external validation, uncertainty quantification, fairness, and clinical utility, and proposed a roadmap.

Selected Technical Projects

Automated Vulnerable-Code Detection – *CodeBERT and distillation*

[Code](#)

- ▶ Built a CodeBERT pipeline classifying code snippets as secure or vulnerable on the DiverseVul dataset; explored distillation to cut inference cost for practical scanning workloads.

Stack Overflow Question Quality Classification – *multi-class NLP pipeline*

[Code](#)

- ▶ Trained an end-to-end classifier over **60,000** questions separating high-quality, community-edited low-quality, and closed low-quality posts.

Bone Fracture Detection – *Vision Transformer (ViT)*

[Code](#)

- ▶ Fine-tuned a pre-trained **ViT-B16** model for binary bone-fracture detection on the **MURA v1.1** radiographic dataset; evaluated generalisation using **5-fold cross-validation**.

Fake-News Detection with SVM Kernels – *classical machine learning*

[Code](#)

- ▶ Compared linear and nonlinear kernels across multiple text representations, reporting accuracy and F1 to separate genuine generalisation from computational cost.

Aid-NSU – Django platform prototype[Code](#)

- Implemented authentication, posting, real-time comments, and chat for a moderated campus platform (prototype; not publicly released).

Professional and Teaching Experience**AI Evaluation and Pilot Task Contributor**

Nov 2025 – Present

MindriftAI

Remote

- Design, review, and validate evaluation scenarios for autonomous AI agents, assessing logical soundness, behavioural alignment, and policy adherence.
- Stress-test multi-step agent flows and improve model behaviour through prompt refinement, curated exemplars, structured feedback, and iterative validation.

Teaching Assistant

Dec 2022 – Oct 2023

North South University

Dhaka, Bangladesh

- Ran tutorials and office hours, prepared course material, designed assignments, and graded for undergraduate computer-science courses.
- Explained technical concepts to mixed-ability cohorts, gave written feedback on student work, and contributed to curriculum-development activities.

Technical Skills

Programming: Python (advanced), SQL, PHP; object-oriented design, data structures, algorithms, scripting.

ML and NLP: PyTorch, Hugging Face Transformers, scikit-learn, spaCy, JAX/Flax (basic), Pandas, NumPy, SciPy; BERT, GPT-family, Mistral, Qwen; classification, regression, segmentation, RAG, LangChain, vector databases.

Model efficiency: Knowledge distillation, 4-bit quantisation, LoRA/QLoRA, gradient checkpointing, adaptive rank scheduling, pruning, efficient evaluation, GPU-memory profiling.

Research methods: Experimental design, controlled baselines, ablation studies, leakage prevention, matched comparison, significance testing, null models, F1/ROC–AUC evaluation, PRISMA literature synthesis.

Systems and tooling: CUDA, Linux, Git/GitHub, reproducible environments, L^AT_EX, n8n, Tableau, Microsoft Office.

Honours, Languages, and Profiles

Honours: Cum Laude, B.S. in Computer Science and Engineering, North South University (2023).

Languages: Bangla (native); English (professional working proficiency : listening C1, reading/writing/speaking B2); Spanish (A1).

Problem solving: Active [LeetCode](#) and [Deep-ML](#) profiles for algorithms, implementation, and applied ML exercises.

References**Professor Dr. Mohammad Ashrafuzzaman Khan**

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North South University

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M.S. thesis and research supervisor

Professor Dr. A. B. M. Obaidul Islam

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Supervising faculty during my Teaching Assistantship at NSU